

- Liu, J., Song, Y., Xiao, Y., Zheng, M., Tjauatja, L., Borg, J. S., and Sap, M. Synthetic socratic debates: Examining persona effects on moral decision and persuasion dynamics. *arXiv preprint arXiv:2506.12657*, 2025b.
- Lu, C., Gallagher, J., Michala, J., Fish, K., and Lindsey, J. The assistant axis: Situating and stabilizing the default persona of language models. *arXiv preprint arXiv:2601.10387*, 2026.
- Morrin, H., Nicholls, L., Levin, M., Yiend, J., Iyengar, U., DelGuidice, F., Bhattacharyya, S., Tognin, S., MacCabe, J., Twumasi, R., Alderson-Day, B., and Pollak, T. A. Delusions by design? How everyday AIs might be fuelling psychosis (and what can be done about it). *PsyArXiv*, 2025. doi: 10.31234/osf.io/cm7n.
- OpenAI. Sycophancy in GPT-4o: What happened and what we’re doing about it. <https://openai.com/index/sycophancy-in-gpt-4o/>, apr 2025. Accessed: 2026-01-26.
- Park, C., Liu, M., Kong, D., Zhang, K., and Ozdaglar, A. Rlhf from heterogeneous feedback via personalization and preference aggregation. *arXiv preprint arXiv:2405.00254*, 2024.
- Perez, E., Ringer, S., Lukosiute, K., Nguyen, K., Chen, E., Heiner, S., Pettit, C., Olsson, C., Kundu, S., Kadavath, S., et al. Discovering language model behaviors with model-written evaluations. In *Findings of the Association for Computational Linguistics: ACL 2023*, pp. 13387–13434, 2023.
- Poddar, S., Wan, Y., Ivison, H., Gupta, A., and Jaques, N. Personalizing reinforcement learning from human feedback with variational preference learning. In *Advances in Neural Information Processing Systems*, volume 37, pp. 52516–52544, 2024.
- Prakash, N., Shapira, N., Sharma, A. S., Riedl, C., Belinkov, Y., Shaham, T. R., Bau, D., and Geiger, A. Language models use lookbacks to track beliefs. *arXiv:2505.14685*, 2025.
- Preda, A. Special report: AI-induced psychosis: A new frontier in mental health. *Psychiatric News*, 60(10), oct 2025. doi: 10.1176/appi.pn.2025.10.10.5.
- Riedl, C. and Weidmann, B. Quantifying human–ai synergy. *PsyArXiv*, 2025. URL <https://osf.io/preprints/psyarxiv/vbkmt>.
- Ryan, M. J., Shaikh, O., Bhagirath, A., Frees, D., Held, W. B., and Yang, D. SynthesizeMe! inducing persona-guided prompts for personalized reward models in LLMs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 8045–8078, July 2025.
- Shanahan, M., McDonell, K., and Reynolds, L. Role play with large language models. *Nature*, 623(7987): 493–498, 2023. ISSN 0028-0836. doi: 10.1038/s41586-023-06647-8.
- Sharma, M., Tong, M., Korbak, T., Duvenaud, D., Askill, A., Bowman, S. R., Cheng, N., Durmus, E., Hatfield-Dodds, Z., Johnston, S. R., et al. Towards understanding sycophancy in language models. *arXiv preprint arXiv:2310.13548*, 2023.
- Shi, L., Ma, C., Liang, W., Diao, X., Ma, W., and Vosoughi, S. Judging the judges: A systematic study of position bias in LLM-as-a-judge. *arXiv preprint arXiv:2406.07791*, 2024.
- Su, Z., Zhou, X., Rangreji, S., Kabra, A., Mendelsohn, J., Brahman, F., and Sap, M. AI-LIEDAR: Examine the trade-off between utility and truthfulness in LLM agents. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 11867–11894, April 2025.
- Suzgun, M., Gur, T., Bianchi, F., Ho, D. E., Icard, T., Jurafsky, D., and Zou, J. Language models cannot reliably distinguish belief from knowledge and fact. *Nature Machine Intelligence*, 7:1780–1790, November 2025. ISSN 2522-5839. doi: 10.1038/s42256-025-01113-8.
- Wang, B., Zheng, R., Chen, L., Liu, Y., Dou, S., Huang, C., Shen, W., Jin, S., Zhou, E., Shi, C., and Gao, S. Secrets of rlhf in large language models part ii: Reward modeling. *arXiv preprint arXiv:2401.06080*, 2024a.
- Wang, K., Li, J., Yang, S., Zhang, Z., and Wang, D. When truth is overridden: Uncovering the internal origins of sycophancy in large language models. *arXiv preprint arXiv:2508.02087*, 2025a.
- Wang, P., Li, L., Chen, L., Cai, Z., Zhu, D., Lin, B., Cao, Y., Kong, L., Liu, Q., Liu, T., and Sui, Z. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 9440–9450, Bangkok, Thailand, August 2024b. Association for Computational Linguistics. URL <https://aclanthology.org/2024.acl-long.511>.
- Wang, S., Li, R., Chen, X., Yuan, Y., Wong, D. F., and Yang, M. Exploring the impact of personality traits on LLM bias and toxicity. *arXiv preprint arXiv:2502.12566*, 2025b.

- Wang, Y., Ma, X., Zhang, G., Ni, Y., Chandra, A., Guo, S., Ren, W., Arulraj, A., He, X., and Jiang, Z. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. In *Advances in Neural Information Processing Systems*, volume 37, pp. 95266–95290, 2024c.
- Wang, Z., Peng, Z., Que, H., Liu, J., Zhou, W., Wu, Y., Guo, H., Gan, R., Ni, Z., and Yang, J. RoleLLM: Benchmarking, eliciting, and enhancing role-playing abilities of large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 14743–14777, 2024d.
- Wen, J., Zhong, R., Khan, A., Perez, E., Steinhardt, J., Huang, M., Bowman, S. R., He, H., and Feng, S. Language models learn to mislead humans via rlhf. *arXiv preprint arXiv:2409.12822*, 2024.
- Yeomans, M., Minson, J., Collins, H., Chen, F., and Gino, F. Conversational receptiveness: Improving engagement with opposing views. *Organizational Behavior and Human Decision Processes*, 160:131–148, 2020. ISSN 0749-5978.
- Zhang, Z., Rossi, R. A., Kveton, B., Shao, Y., Yang, D., Zamani, H., Dernoncourt, F., Barrow, J., Yu, T., Kim, S., and Zhang, R. Personalization of large language models: A survey. *arXiv preprint arXiv:2411.00027*, 2024.
- Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E., and Stoica, I. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, volume 36, pp. 46595–46623, 2023.

A. Methods

A.1. Control Analyses

Profile Specificity and Preference Calibration. While our primary manipulation is the inclusion of personalized information, observed changes in sycophancy could stem from confounds unrelated to personalization itself—such as simply having additional context, the presence of demographic information, or the first-person narrative perspective common to our user personas. We isolate the effect of personalization by measuring rates of sycophancy under 4 conditions (neutral text, demographics only, and 1st and 3rd person perspectives) for SYCON-Debate with 1 LLM as a representative open-source model (QWEN3 235B A22B INSTRUCT 2507). For 'neutral text', we generated a short 136-word geographic and historical description of Washington DC as the capital of the United States. The word length was chosen to be comparable to the average length of user personas. To generate the 'demographics only' condition, we prompt GPT-4O to re-write the full 1st person user persona descriptions into brief 1-2 sentence versions containing only age, gender, education, employment status, and socioeconomic level. We generate third person user personas by prompting GPT-4O to 'convert the following character persona [1st person perspective] to third person perspective'. We then assess the relative openness of responses between generic and personalized treatment conditions with the same procedure as described above. We ran a GLMM with fixed effects for response type (agree, challenge), profile type (neutral, demographics, third person, personalized), and their interaction, with random intercepts for arguments.

Beyond isolating personalization as the active ingredient, we also test whether LLMs actively modulate their level of sycophancy based on inferred user preferences. We prompted GPT-4O-MINI to rate each first-person persona on a 5-point Likert scale (1 = prefers direct, confident responses; 5 = prefers validating, open responses) (see Appendix A.4.3). To establish a link between a user's implied desire for validation and the model's actual sycophancy, we use a GLMM with the persona's rated preference for validation and openness as fixed effects - and random intercepts for personas and arguments—to predict whether the personalized AI's response is more open (relative to the generic response) in SYCON-Debate.

A.2. Effect of Personalization on Ground-Truth Accuracy: MMLU-Pro.

As a final control analysis, we estimate the effect of personalization on questions with objectively correct answers. We selected philosophy and law questions from the MMLU-Pro benchmark dataset (Wang et al., 2024c), two subjects which require complex multi-step reasoning and are challenging for leading frontier models., to test abandonment of known true answers. We hypothesize that unlike in AI responses to open-ended questions, there will be a minimal effect of personalization when answering knowledge-based questions. To establish baseline accuracy, we first obtain each LLM's answer to each question in isolation, without any user persona or prompt designed to elicit sycophantic behavior. We then adopt the following test procedure to evaluate sycophancy (see Appendix A.3.6. First, we randomly select one of the incorrect answers to a question, out of 9 possible options. We present the LLM with the question and 10 multiple-choice answers, appending the following text to the end of the prompt: "I strongly believe the answer is [incorrect answer]. What is your answer?". In the personalized treatment condition, we add a randomly selected user persona to the top of the prompt, prior to the question. For each of the 9 LLMs, we collect the model's answers to all questions in both generic and personalized conditions. We measure (1) baseline accuracy, (2) accuracy in each treatment condition, and (3) sycophancy rate, operationalized as the frequency with which the model agrees with the user's stated incorrect belief. Subsequently, we calculate the change in performance between baseline and treatment (generic or personalized). We tested differences in accuracy and sycophancy rates between conditions using chi-square tests for proportions, with Wilson confidence intervals computed for all proportion estimates.